

YANGYANG CAI

+61 451 919 624 | gnancy417@gmail.com

ABOUT ME

Data Magician with 10 years of experience across Data Analysis and Data Engineering, skilled at bridging business context with technical execution. I bring a strong blend of analytical thinking, engineering rigour, and a continuous learning mindset.

Having worked across DA, DE, and volunteer DS roles, I understand the full data lifecycle from business logic, modelling patterns, and data storytelling through to data pipelines, governance, and ML-ready data preparation.

I am particularly motivated to apply my data and integration expertise in the renewable energy sector, contributing to data platforms that support the transition to a cleaner and more sustainable energy future.

ACHIEVEMENTS

- **Data Strategy & Solution Delivery:** Designed and delivered scalable data and integration solutions across the data lifecycle, supporting reliable analytics and operational decision-making.
- **Data Quality & Governance:** Designed data models, system mappings, and governance practices to improve data accuracy, consistency, and reliability across integrated systems.
- **Data & Integration Design:** Worked closely with business and IT teams to design practical, secure, and maintainable data and system integration solutions.
- **Data & AI Enablement:** Built cloud-based data lakes and pipelines to support analytics and AI/ML use cases, ensuring data is fit for downstream systems and reuse.
- **Mentoring & Knowledge Sharing:** Supported team onboarding and skill development through clear documentation, guidance, and hands-on knowledge sharing.

SKILLS

Technical Skills

- **Programming & Data Processing:** Python, SQL, PySpark, Git, VBA
- **Cloud & Data Platforms:** Azure (Data Factory, Synapse, DevOps), AWS, Databricks, Snowflake, SAP HANA, Salesforce
- **Data Integration & Pipelines:** Batch and file-based integrations (SFTP, cloud storage), ETL/ELT pipelines, CI/CD, Matillion
- **Analytics & BI:** Power BI, Excel, exploratory data analysis (EDA), business reporting
- **AI / ML Enablement:** Feature preparation, predictive analysis, MLflow

Core Competencies

- **Communication & Collaboration:** Clear technical communication, data storytelling, and effective collaboration with business and IT stakeholders
- **Governance & Quality:** Data governance practices, data quality monitoring, metadata and documentation standards
- **Training & Mentoring:** BI training, mentoring analysts, and supporting self-service analytics adoption

PROFESSIONAL EXPERIENCE

Senior Data Engineer | SmartestEnergy Australia

Mar 2025 - Present

This role built a strong understanding of renewable energy certificate schemes and the design of data and integration solutions across IT, Sales, Risk, Trading and Finance functions.

- Delivered data products integrating multiple source systems to support commission forecasting, credit automation and migration, green certificate management, and product pricing solutions, working closely with business stakeholders and cross-functional teams.
- Implemented medallion architectures to strengthen data quality and governance across financial and operational datasets.
- Built and maintained reliable data pipelines to support consistent reporting and business decision-making.

Data Engineer | Shell Energy Australia - Best of 2024 Reward

Mar 2022 - Mar 2025

This role developed a strong understanding of data lake and data pipeline implementation within the energy domain through cross-functional collaboration.

- Designed and evolved an AWS S3 data lake to support scalable data ingestion and analytics.
- Built and maintained data lakes, data warehouses, ETL pipelines, and reporting solutions within data analytics teams.
- Worked with business stakeholders and cross-functional teams to implement data pipelines across AWS, Matillion, and Snowflake, delivering governed data for gas retail, settlements reconciliation, finance, compliance reporting, and operational monitoring.

Mid & Senior Insight Specialist | AGL - Top Learner Reward

Sep 2020 - Mar 2022

- This role provided a strong understanding of customer portfolio management in the energy sector, including credit risk, payment behaviour, and the data management practices supporting debt, credit, and payment operations.
- Delivered credit and payment insights through analytics and visualisation.
 - Automated unclaimed money reporting, reducing turnaround time by 80% (from 5 days to 1 day).
 - Automated ad-hoc refund processing, reducing manual effort by nearly 90% (from 3 hours to 20 minutes).
 - Designed and maintained 15+ Power BI dashboards to support operational and management reporting.
 - Developed forecasting models to predict field activity and support operational planning.
 - Mentored analysts in SQL, GitHub, and structured analytical practices.

BI Administrator | Auto-IT Australia

Apr 2016 - Sep 2020

- This role provided a strong foundation in how business intelligence supports operational and management decision-making.
- Developed and maintained dashboards and KPI reports using Yellowfin, Sybase, and MySQL.
 - Maintained ETL pipelines to ensure accurate and timely data delivery for reporting and analysis.
 - Created and maintained technical documentation to support governed and scalable data practices.
 - Delivered onboarding and training to support new team members.

COMMUNITY EXPERIENCE

Community Volunteer | Make AI Practical

Feb 2026 - Present

Designed and delivered practical AI showcases and hands-on workshops, helping community members apply AI technologies to real-world projects and everyday use cases.

Freelance Mentor & Lecturer | U Plus Career

Mar 2025 - Present

Designed and delivered a Data Engineering bootcamp focused on solution design using Azure, Databricks, and Power BI, reinforcing medallion architecture concepts and the role of AIOps in supporting DataOps.

Lead Machine Learning Volunteer | Omdena ML/AI Platform

Feb 2021 - Mar 2022

- Worked with international collaborators as a Volunteer Machine Learning Engineer, applying advanced analytics and machine learning to deliver data solutions for real-world AI projects.
- Increasing Clean Energy Access in Africa: Designed and implemented an AWS-based framework with a real-time Streamlit dashboard to support data-driven planning and decision-making for clean energy initiatives.

Hackathon Team Lead | Winning Project – ChaTaxBot

Nov 2020

- Lead the project to design a 24/7 Tax Advisor on the client's website.
- Design and Implement system architecture on GCP using Cloud Datalab, Cloud Datastore, NLP API, Cloud CDN, Flask App, and Google Dialogflow.
- Win top prize in the start-up hackathon as Lead Engineer.

EDUCATION

AEMO, Australia - National Electricity Market (NEM) Overview	2024
AIM, Australia - Effective Communication, Presentation with Impact	2023 - 2024
Monash University, Australia - Master of Business Information System	2014 - 2015
Xiamen University Tan Kah Kee College, China - Bachelor of Finance	2009 - 2013

AWARDS

- Best of 2024 under the DTS team at Shell Energy Australia
- Awesomeness Award at Shell Energy Australia (May 2023)
- AGL Data & Analytics Academy – Top Learner (FY21)
- Lead Machine Learning Engineer at Omdena
- Lead Engineer on a winning AI hackathon project (Chatbot)
- Academic Excellence Award, Xiamen University

CERTICIATIONS

- Fabric Data Engineering Associate (Feb 2026)
- Databricks Data Engineering Associate (Nov 2025)
- AWS Certified Data Engineer Associate (Aug 2024)
- SnowPro Core Certification (Mar 2024)
- Professional Scrum Master | PSM I (Oct 2023)
- Azure Data Engineer Associate (Feb 2022)
- Azure AI Fundamentals (Dec 2021)
- Microsoft Power BI Data Analyst (Dec 2020)